

[T]-Theory: Geophysics

P12 · Collective Field · For: The Seismologist

Seismic Memory Propagator — Earth's elastic Green's function as a USF instance.

Scale 10–11 identification:

- Scale 10: Earth's elastic Green's function — seismic wave propagation
- Scale 11: Seismic Memory Propagator — global free oscillations
- Same equation: $(\nabla^2 + k^2)G = \delta$ with crustal parameters
- Normal modes = poles of G_{10} at eigenfrequencies ω_n

Cross-scale correspondence:

- Seismic P -wave $\leftrightarrow$ longitudinal field mode
- Seismic S -wave $\leftrightarrow$ transverse field mode
- Earth's free oscillations $\leftrightarrow$ global field normal modes
- Mantle convection $\leftrightarrow$ thermodynamic propagator (scale 12)

I · The Seismic Memory Propagator

The elastic Green's function for seismic waves in a stratified Earth:

$$G_{ij}^E(\mathbf{r}, t; \mathbf{r}', t') = \sum_n \frac{u_n^i(\mathbf{r}) u_n^{*j}(\mathbf{r}')}{\omega_n^2 - \omega^2 - 2i\omega\gamma_n} e^{-i\omega_n(t-t')}$$

This is the **resolvent** of the elastic wave operator — mathematically identical to the Clinical Operator Propagator (P10) under relabelling. One equation; different substrate.

II · Normal Modes and Memory

Earth's normal modes (free oscillations ${}_0S_2, {}_0T_2, \dots$) are exactly the poles of G^E . Their frequencies encode the Earth's interior structure — the planet's somatic memory.

Post-earthquake ring-down $\equiv$ memory kernel decay:

$$G^E(t) \sim \sum_n A_n e^{-t/\tau_n} \cos(\omega_n t + \phi_n)$$

This is the same form as $K(\tau) = K_0 e^{-\tau/\tau_m}$ from somatic memory (scale 7) — the *Seismic Memory Propagator*.

III · The USF at Geological Scale

- Scale 10: local seismic propagation — fault zone to fault zone
- Scale 11: global free oscillations — the whole Earth as a resonator
- Scale 12: mantle convection and solar wind — the thermodynamic propagator
- Cross-scale link: USF wave speed $v_s(\sigma)$ interpolates from neural to seismic

IV · Geophysical Test of the USF

Prediction: the ratio of seismic P to S wave speeds should equal the ratio of corresponding USF field modes at scale 10.

$$\frac{v_P}{v_S} = \sqrt{\frac{\lambda + 2\mu}{\mu}} \approx \frac{k_P}{k_S}$$

This is a direct numerical test of the scale-10 G-identification. Data: global seismograph networks (IRIS/GSN).

G-ID: *Seismic Memory Propagator* — elastic Green's function for crustal wave propagation. ORCID: 0009-0007-2194-0850 · CC BY 4.0 · Zurich 2026